# Lecture - 13

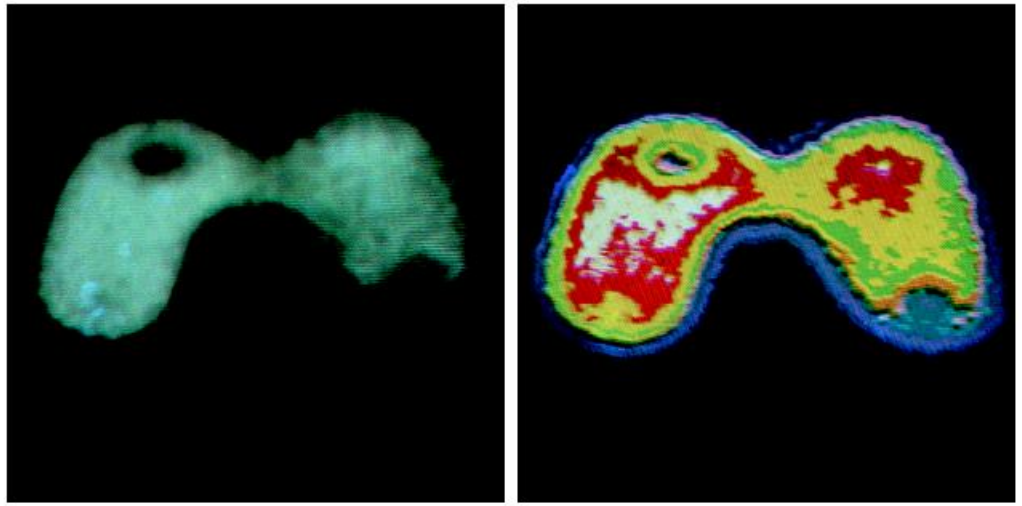
# Chapter 6

# COLOR IMAGE PROCESSING

# Color Fundamentals

- Why use color in image processing?
  - Color is a powerful descriptor
    - Object identification and extraction
    - E.g. Face detection using skin colors

  - Humans can discern thousands of color shades and intensities
    - Human discern only two dozen shades of grays



# Color Fundamentals

- Two category of color image processing
  - Full color processing
    - Images are acquired from full-color sensor or equipments
  - Pseudo-color processing
    - In the past decades, color sensors and processing hardware were not available
    - Colors were assigned to a range of monochrome intensities

# Color Fundamentals

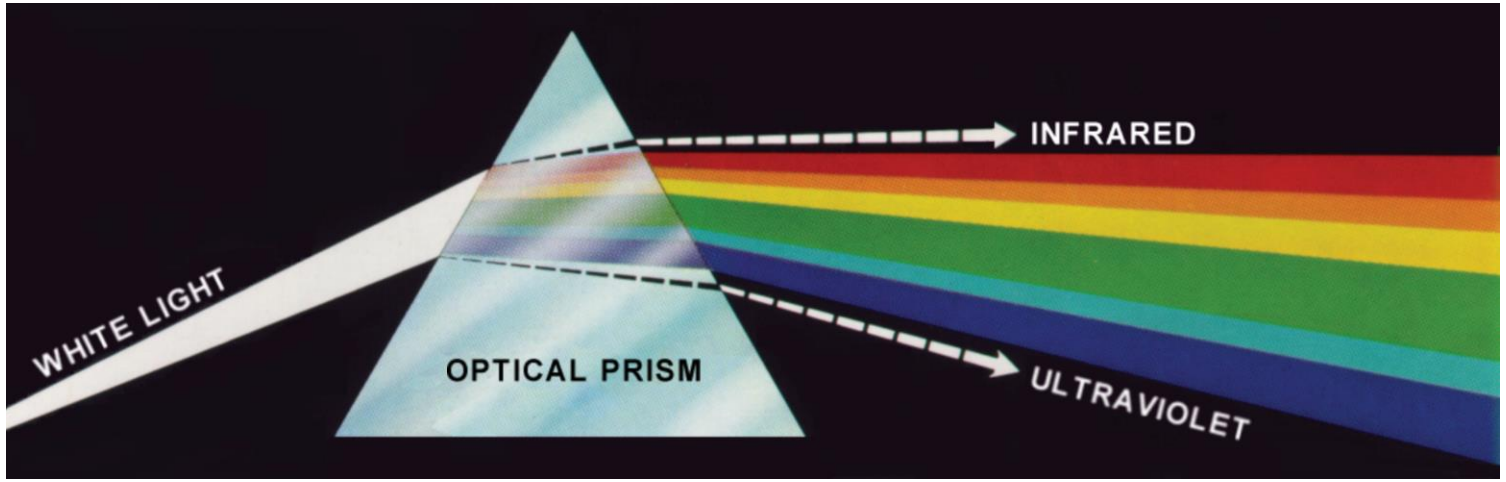



**FIGURE 6.1** Color spectrum seen by passing white light through a prism. (Courtesy of the General Electric Co., Lighting Division.)

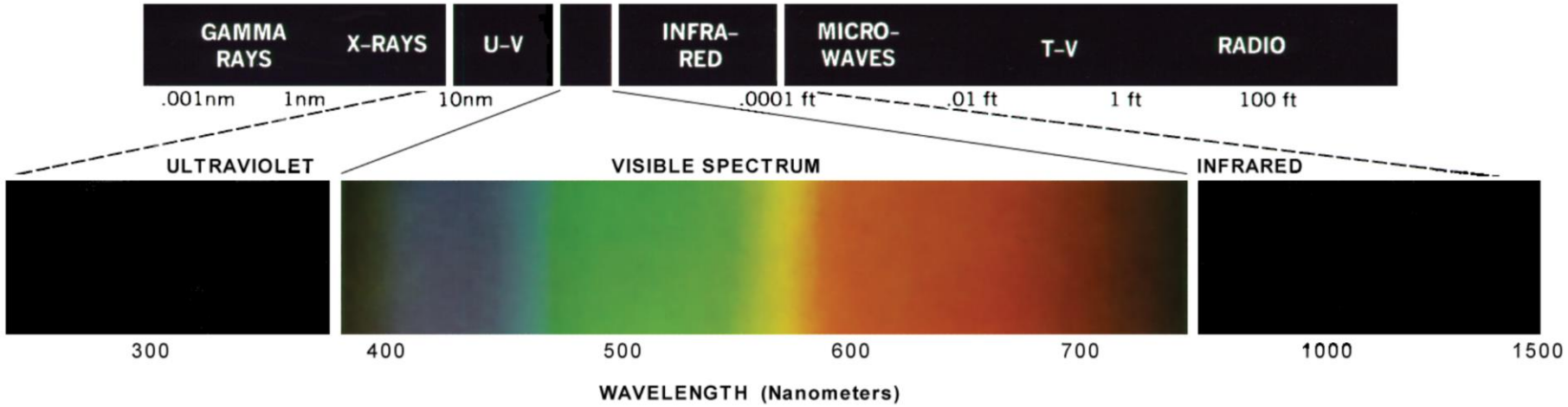



**FIGURE 6.2** Wavelengths comprising the visible range of the electromagnetic spectrum. (Courtesy of the General Electric Co., Lighting Division.)

# Color Fundamentals

6~7M Cones are the sensors in the eye

3 principal sensing categories in eyes: Red light 65%, green light 33%, and blue light 2%

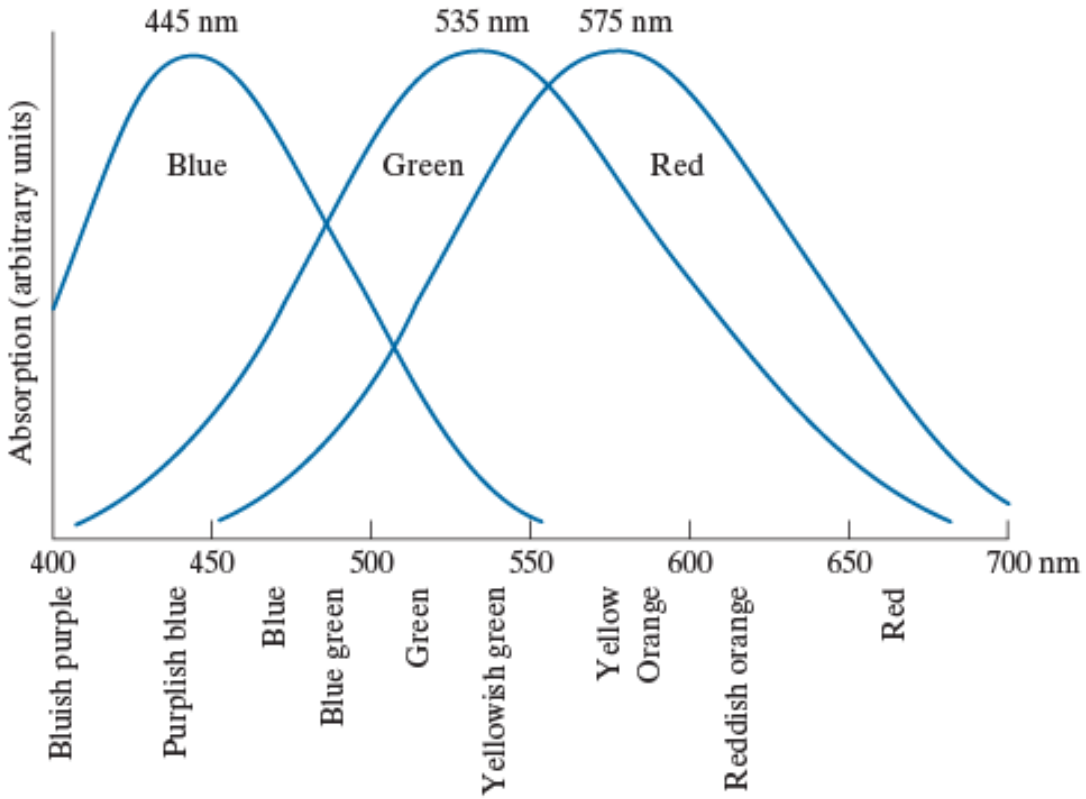
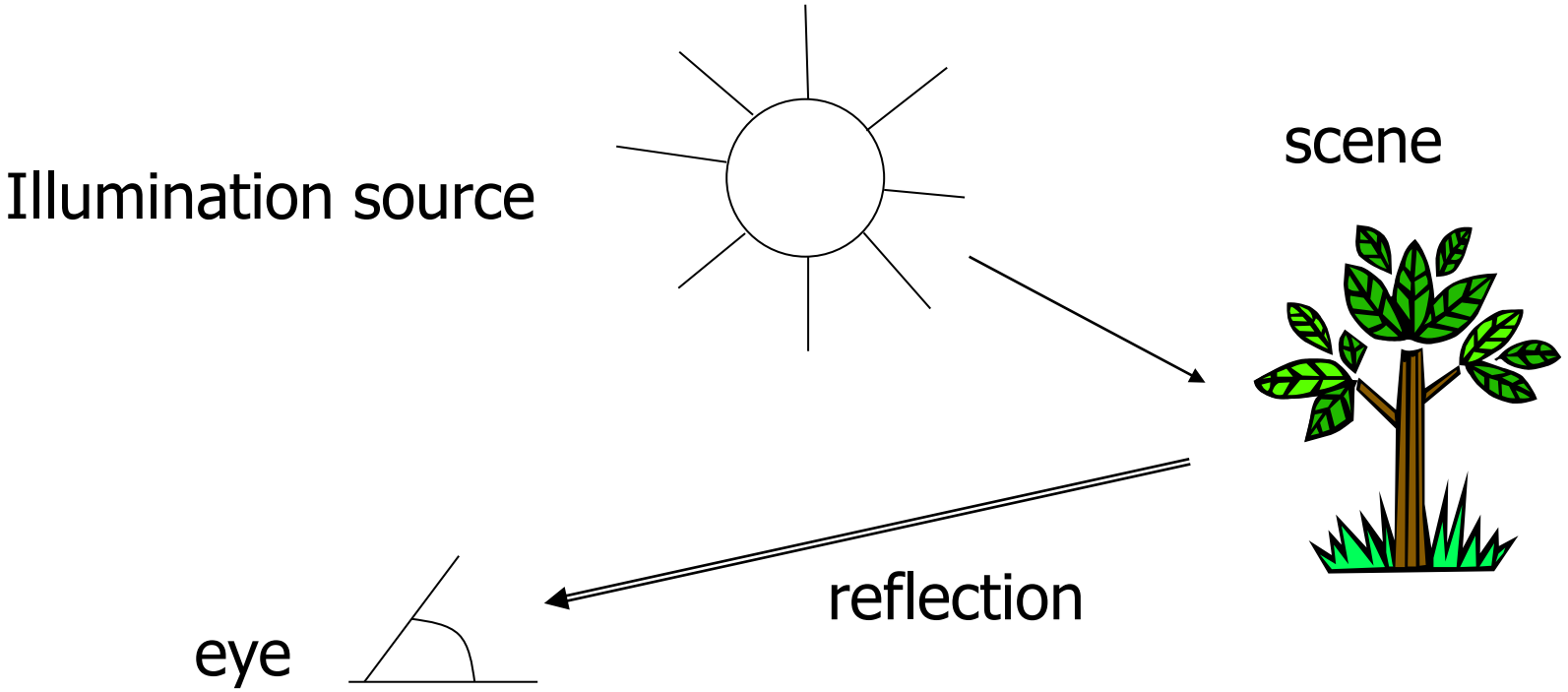



**FIGURE 6.3** Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.

# Color Fundamentals

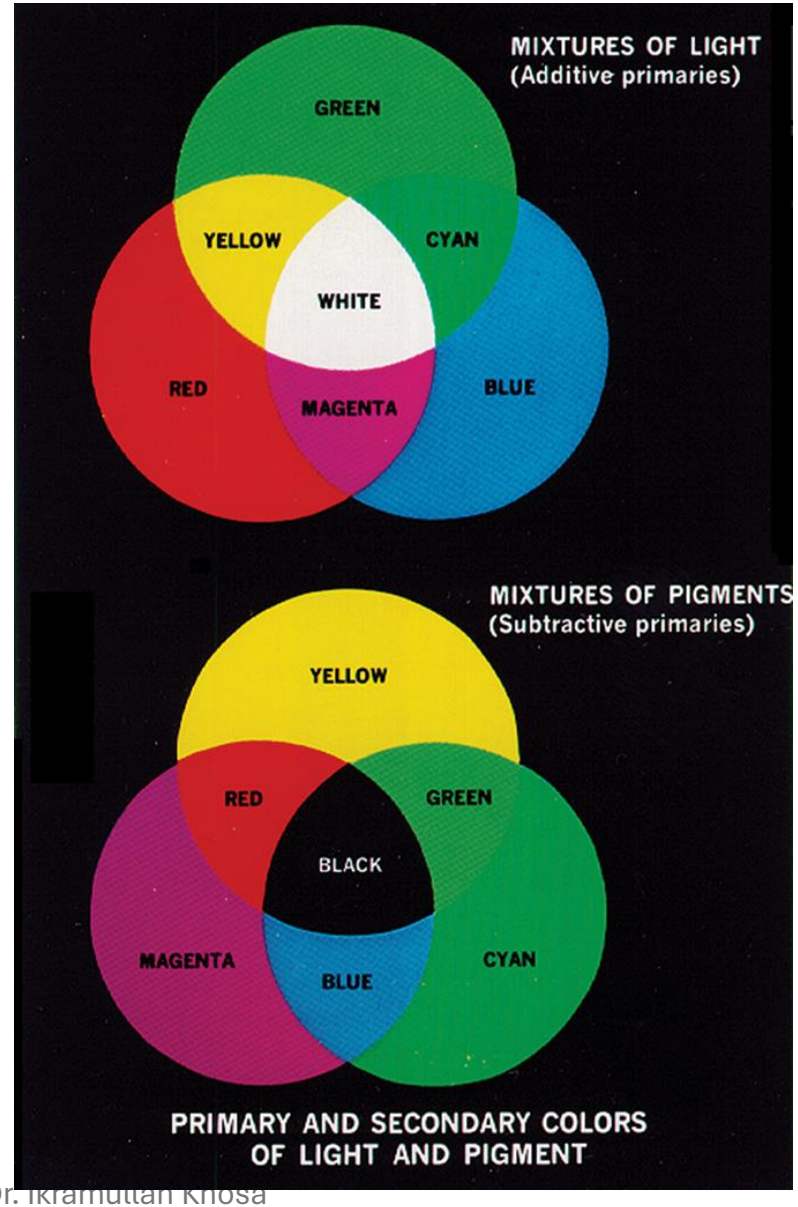
- The color that human perceive in an object = the light reflected from the object



# Primary colors of light vs. primary colors of pigments

- Primary color of pigments
  - Color that subtracts or absorbs a primary color of light and reflects or transmits the other two

| | | | |
|---|---|---|---|
| Color of light: | R | G | B |
| | ↓ | ↓ | ↓ |
| Color of pigments: | absorb R<br>Cyan | absorb G<br>Magenta | absorb B<br>Yellow |




a
b

**FIGURE 6.4** Primary and secondary colors of light and pigments. (Courtesy of the General Electric Co., Lighting Division.)

# Chromaticity Diagram

Normalized tristimulus values

$$x = \frac{X}{X+Y+Z}$$

$$y = \frac{Y}{X+Y+Z}$$

$$z = \frac{Z}{X+Y+Z}$$

x+y+z=1

x, y (chromaticity coordinate) is enough to describe all colors
z = 1 – (x+y)

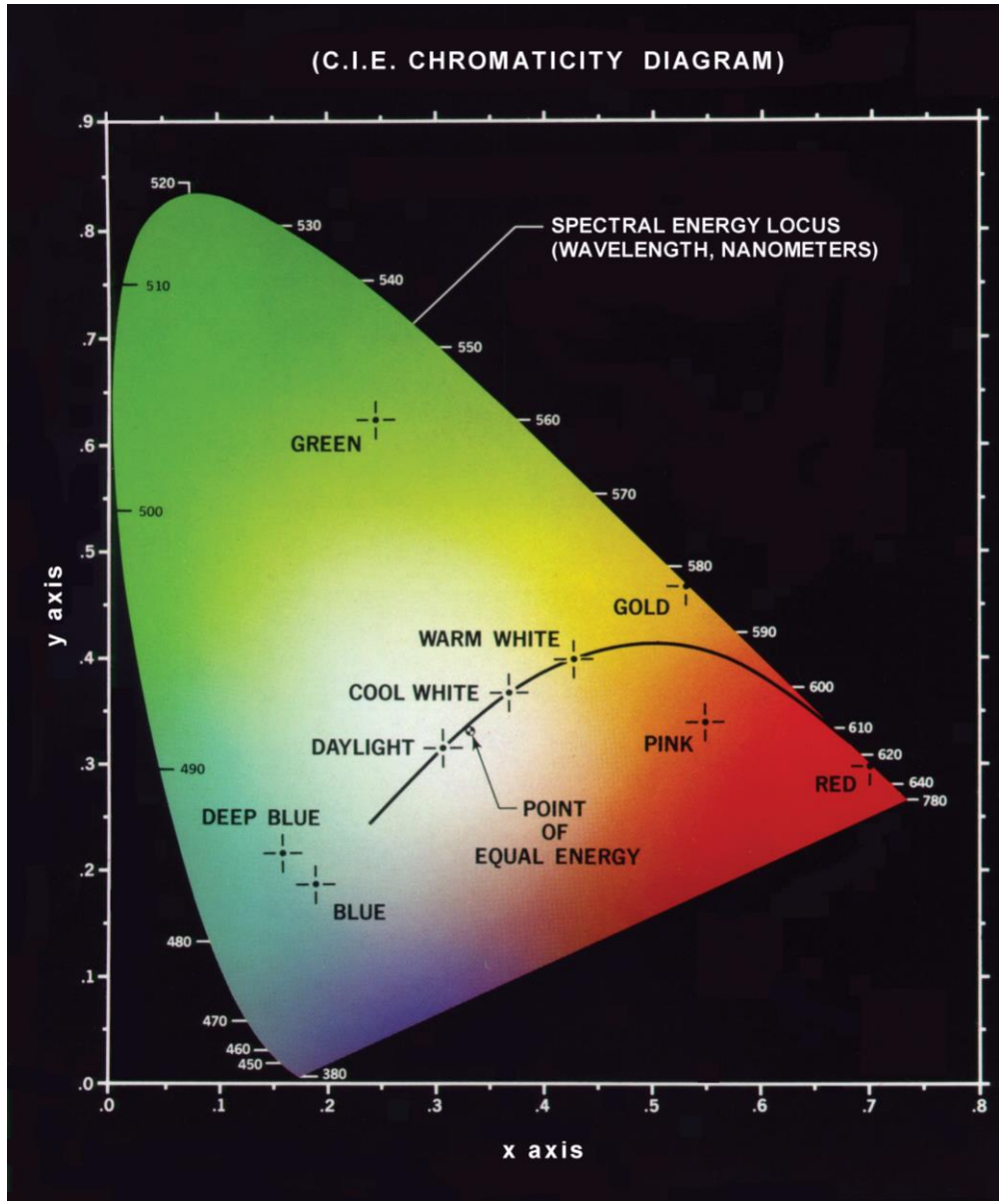



**FIGURE 6.5** The CIE chromaticity diagram. (Courtesy of the General Electric Co., Lighting Division.)

# Chromaticity Diagram

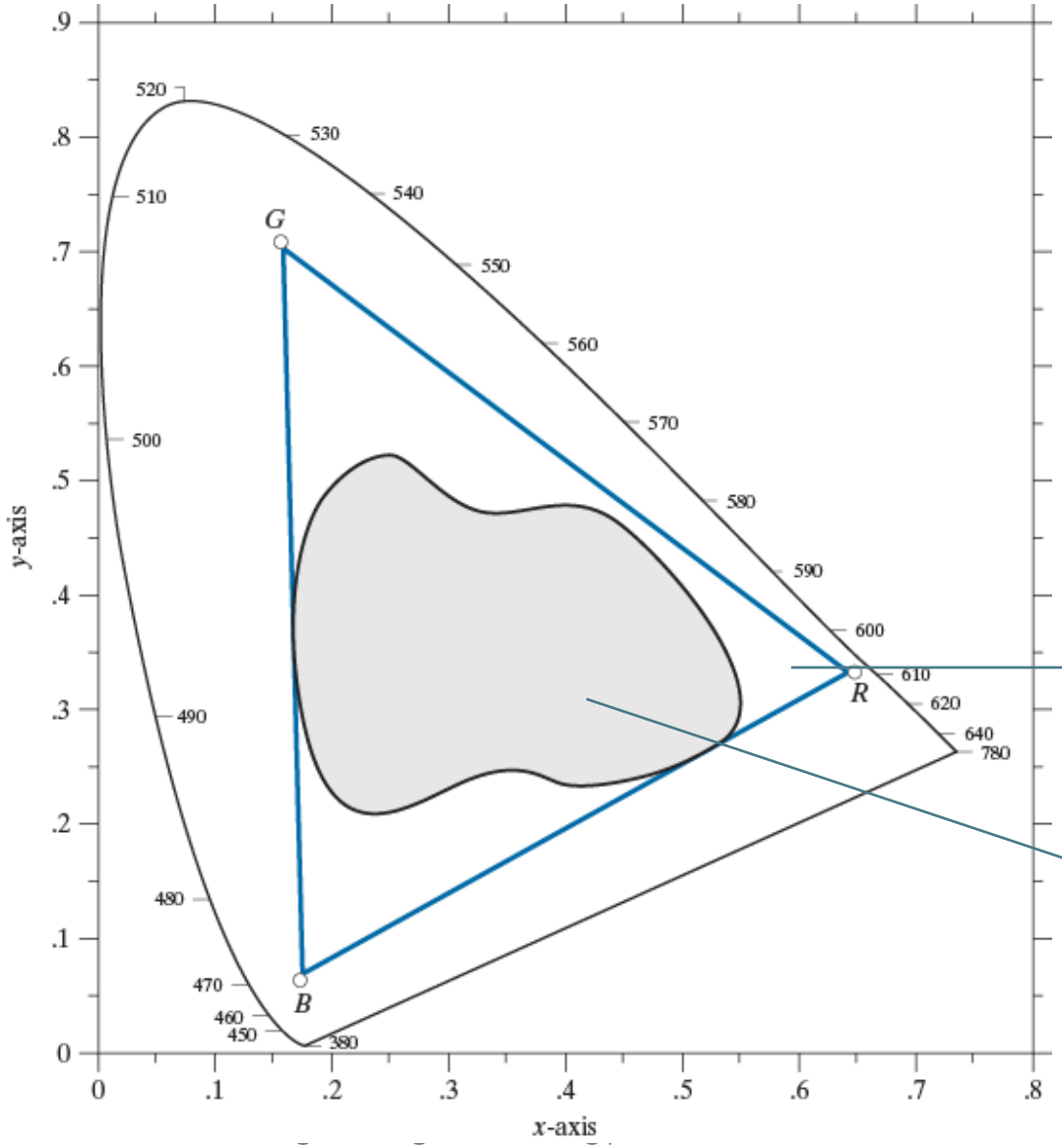



**FIGURE 6.6** Illustrative color gamut of color monitors (triangle) and color printing devices (shaded region).

RGB gamut of monitors

Color gamut of printers

# Color Model

- Color model, color space, color system
  - Specify colors in a standard way
  - A coordinate system that each color is represented by a single point

- RGB model
- CYM model
- CYMK model

} Suitable for display/hardware

- HSI model - Software processing (match the human description)

# The RGB Color Model

- Pixel depth: the number of bits used to represent each pixel in RGB space

- Full-color image: 24-bit RGB color image
  - (R, G, B) = (8 bits, 8 bits, 8 bits)

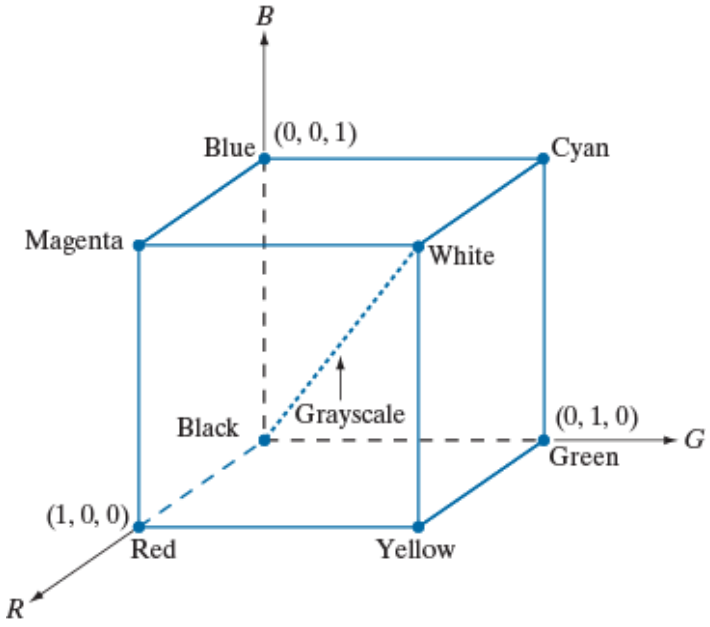



**FIGURE 6.7** Schematic of the RGB color cube. Points along the main diagonal have gray values, from black at the origin to white at point (1, 1, 1).

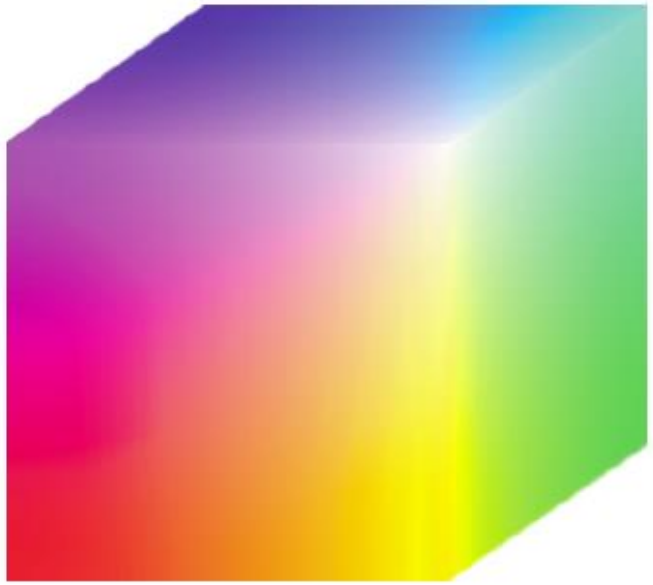
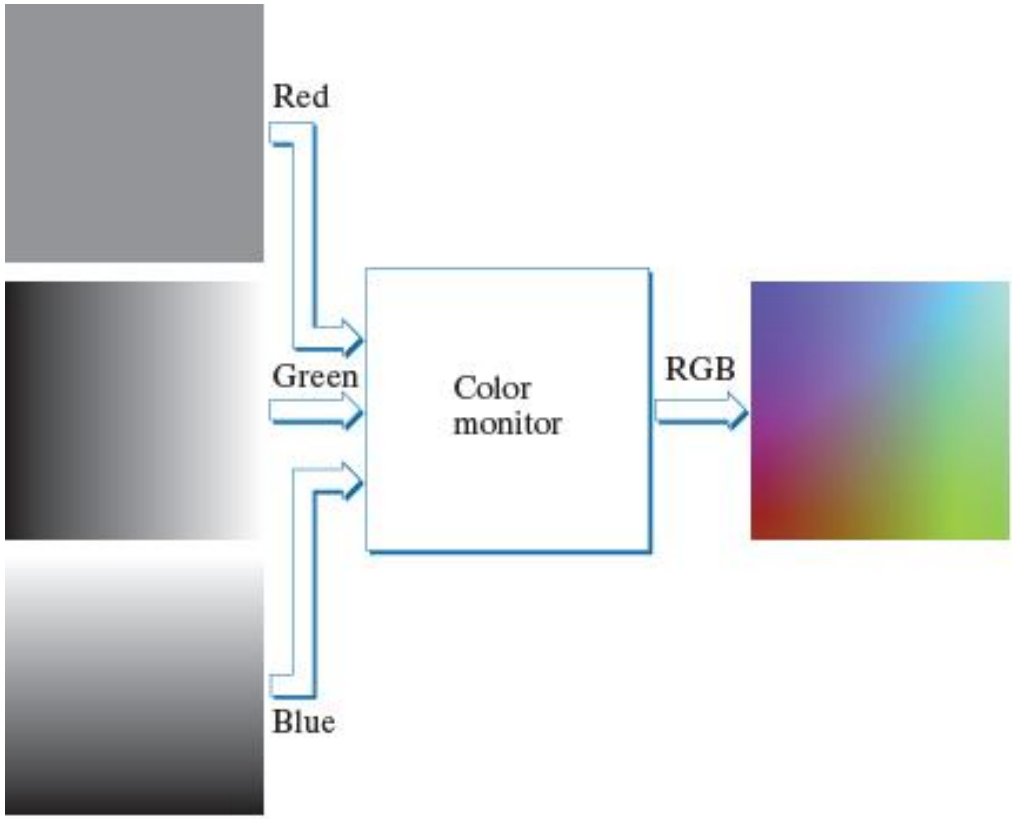
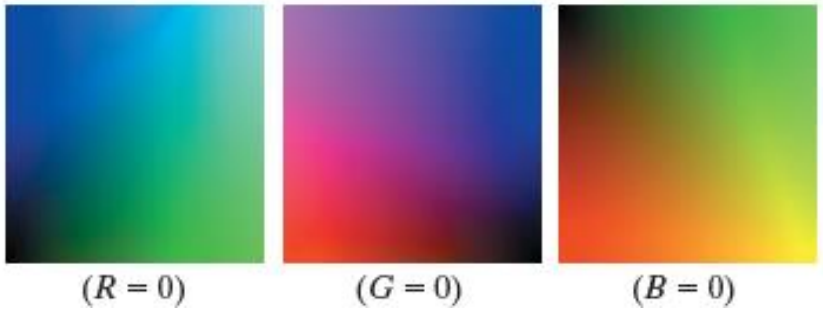


**FIGURE 6.8** A 24-bit RGB color cube.

# The RGB Color Model




a
b

**FIGURE 6.9** (a) Generating the RGB image of the cross-sectional color plane (127, G, B). (b) The three hidden surface planes in the color cube of Fig. 6.8.

# Safe RGB Colors

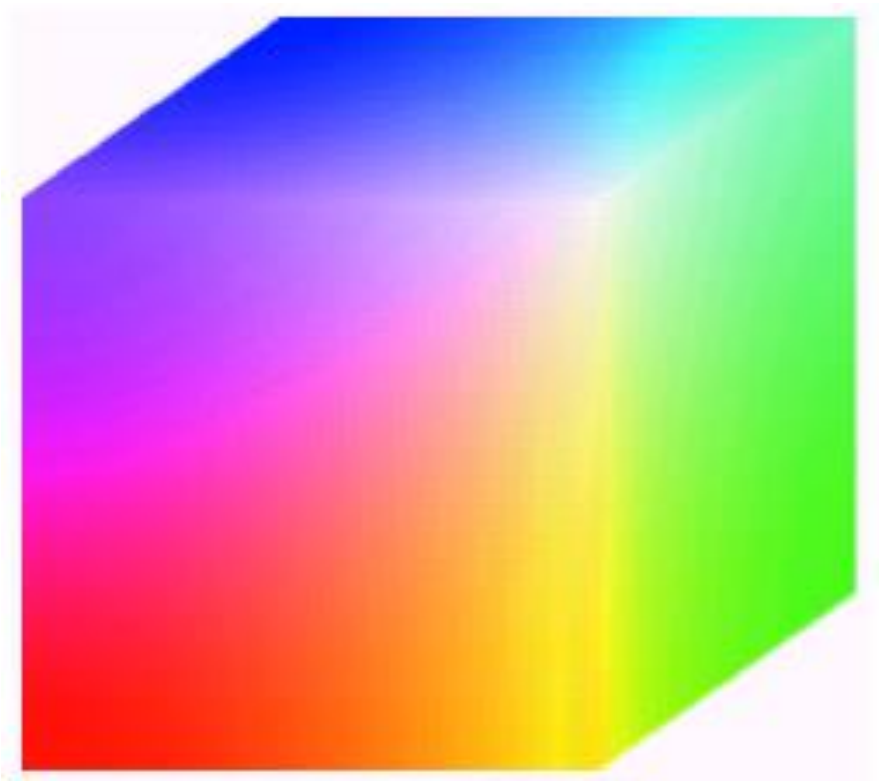
- Subset of colors is enough for some application
- Safe RGB colors (safe Web colors, safe browser colors)

| Number System | Color Equivalents | | | | | |
|---|---|---|---|---|---|---|
| Hex | 00 | 33 | 66 | 99 | CC | FF |
| Decimal | 0 | 51 | 102 | 153 | 204 | 255 |

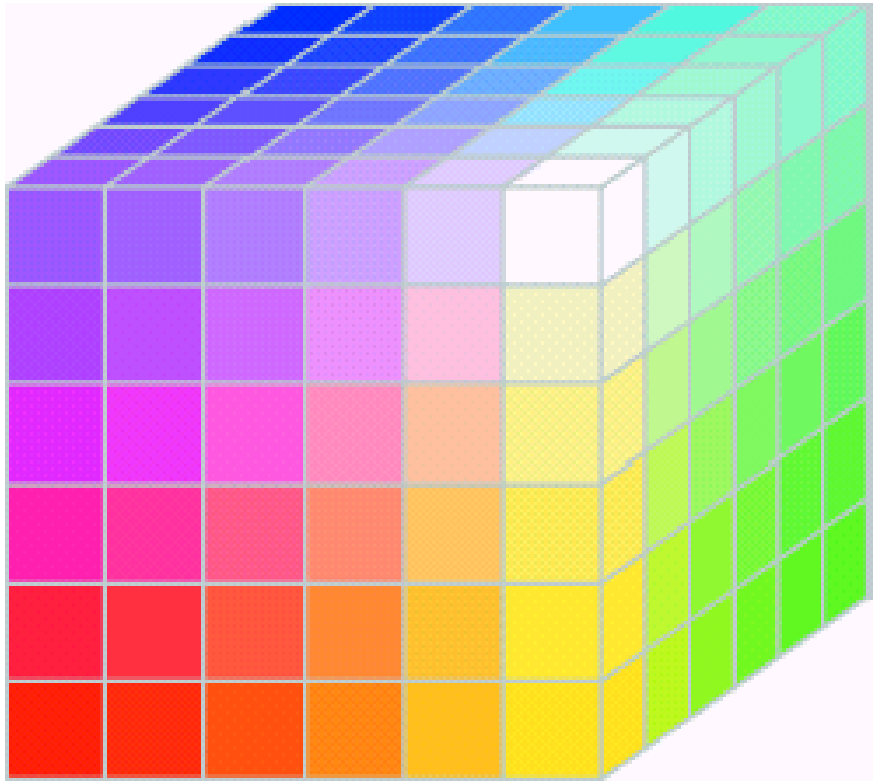
**TABLE 6.1** Valid values of each RGB component in a safe color.

$(6)^3 = 216$

# Safe RGB Colors



Full color cube

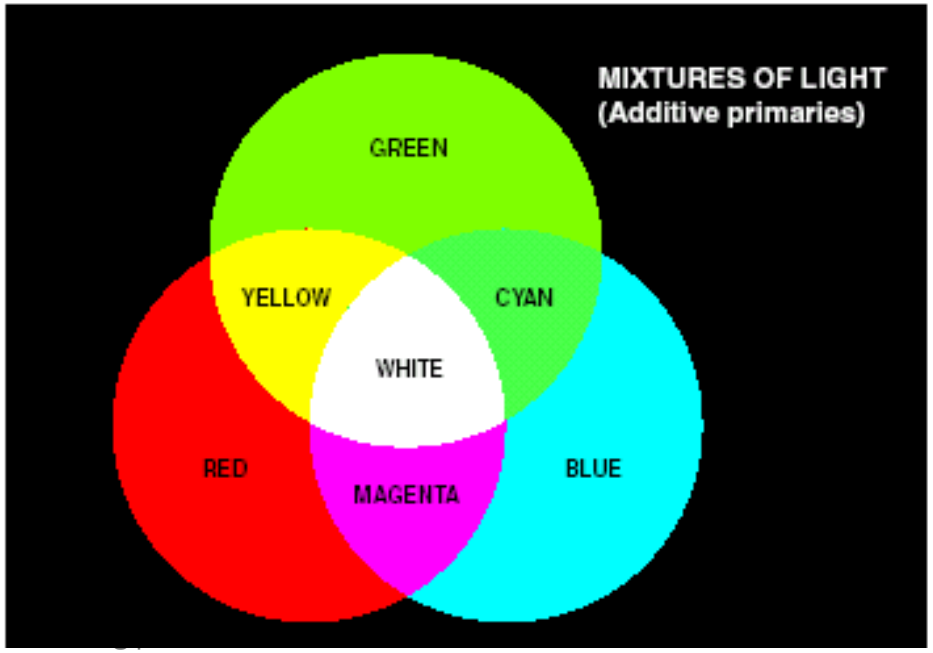


Safe color cube

# CMY (+Black = CMYK) Color Model

- CMY: secondary colors of light, or primary colors of pigments
- Used to generate hardcopy output

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$



# HSI Color Model

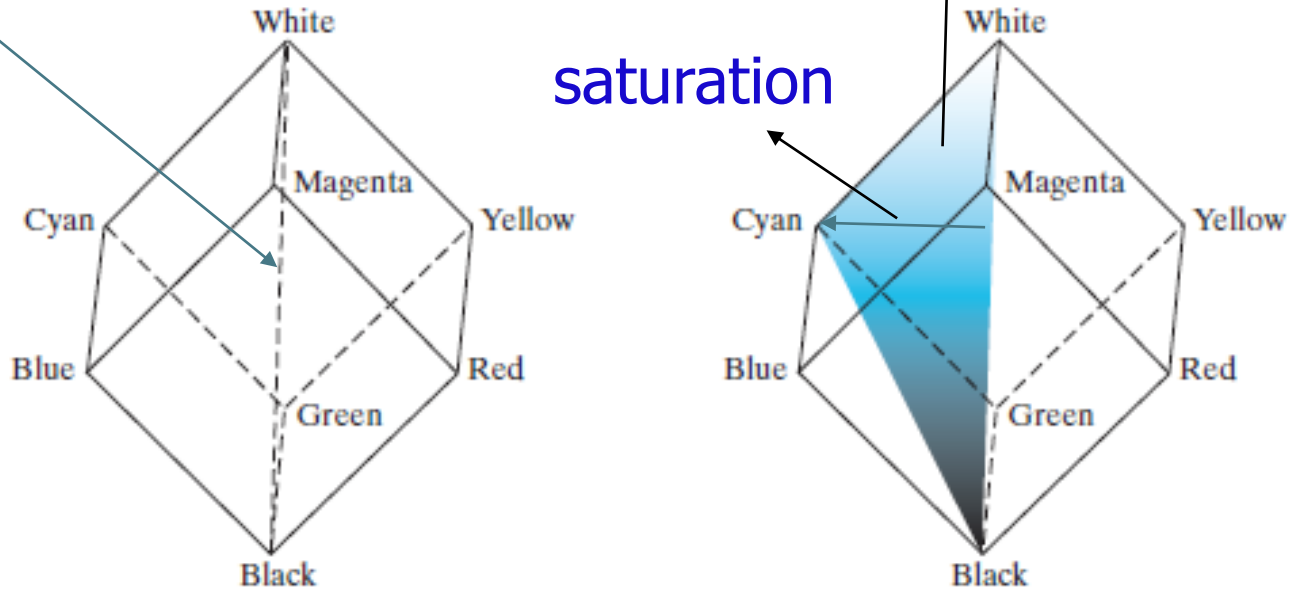
- Will you describe a color using its R, G, B components?
- Human describe a color by its hue, saturation, and brightness
  - Hue : color attribute
  - Saturation: purity of color (white->0, primary color->1)
  - Brightness: achromatic notion of intensity

# HSI Color Model

Intensity line

Colors on this triangle
Have the same hue
(Iso-hue triangle)

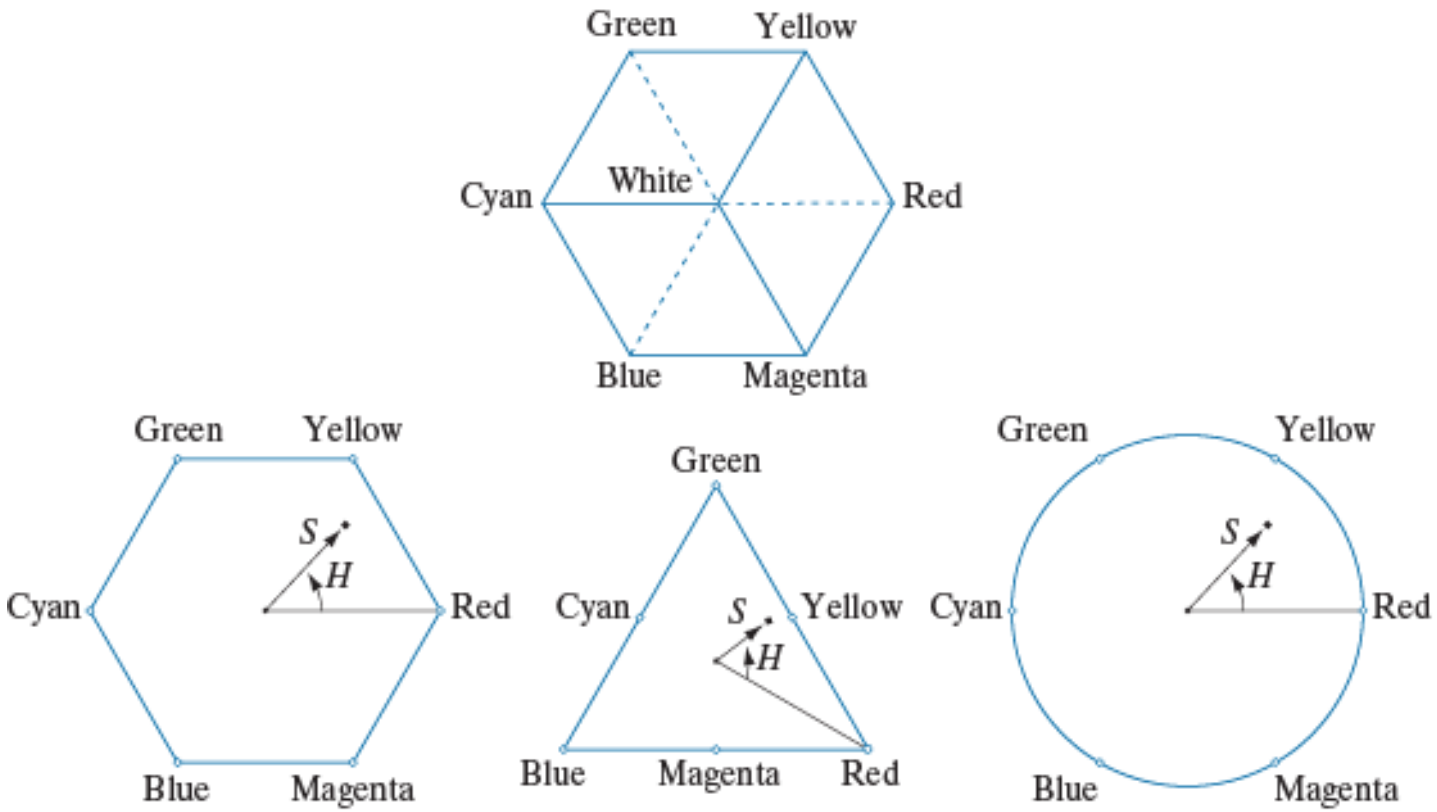
saturation




a b

**FIGURE 6.10** Conceptual relationships between the RGB and HSI color models.

# HSI Color Model




a
b c d

**FIGURE 6.11** Hue and saturation in the HSI color model. The dot is any color point. The angle from the red axis gives the hue. The length of the vector is the saturation. The intensity of all colors in any of these planes is given by the position of the plane on the vertical intensity axis.

# HSI Color Model

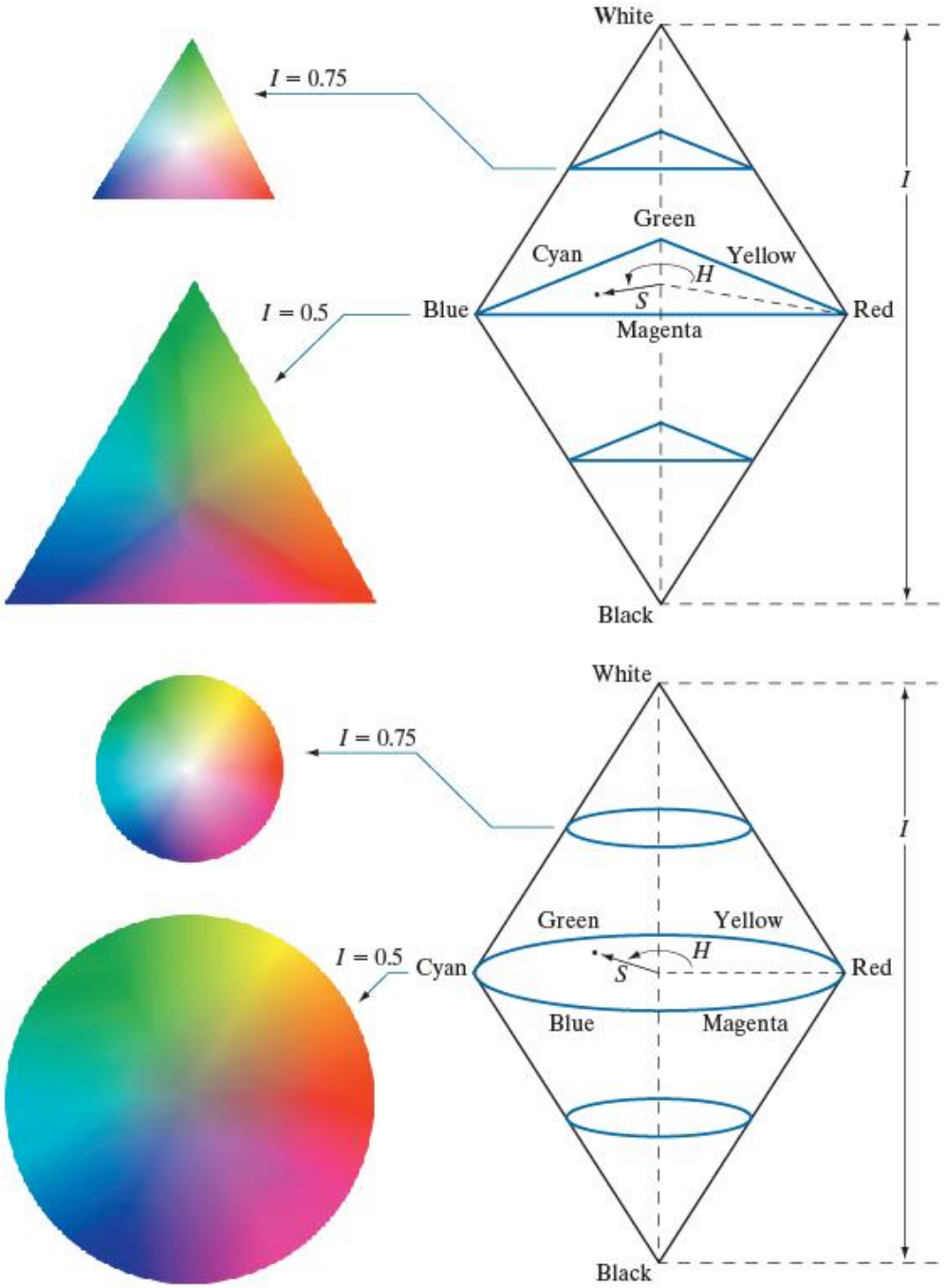
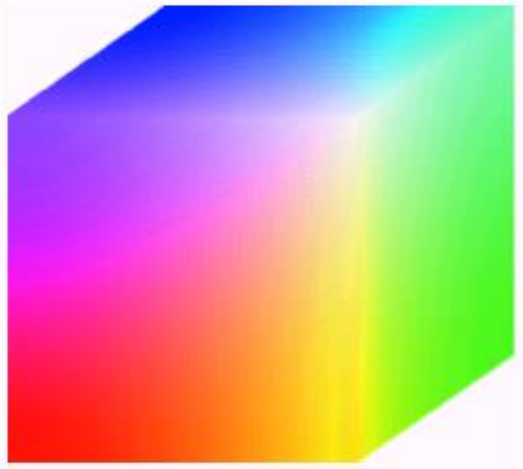
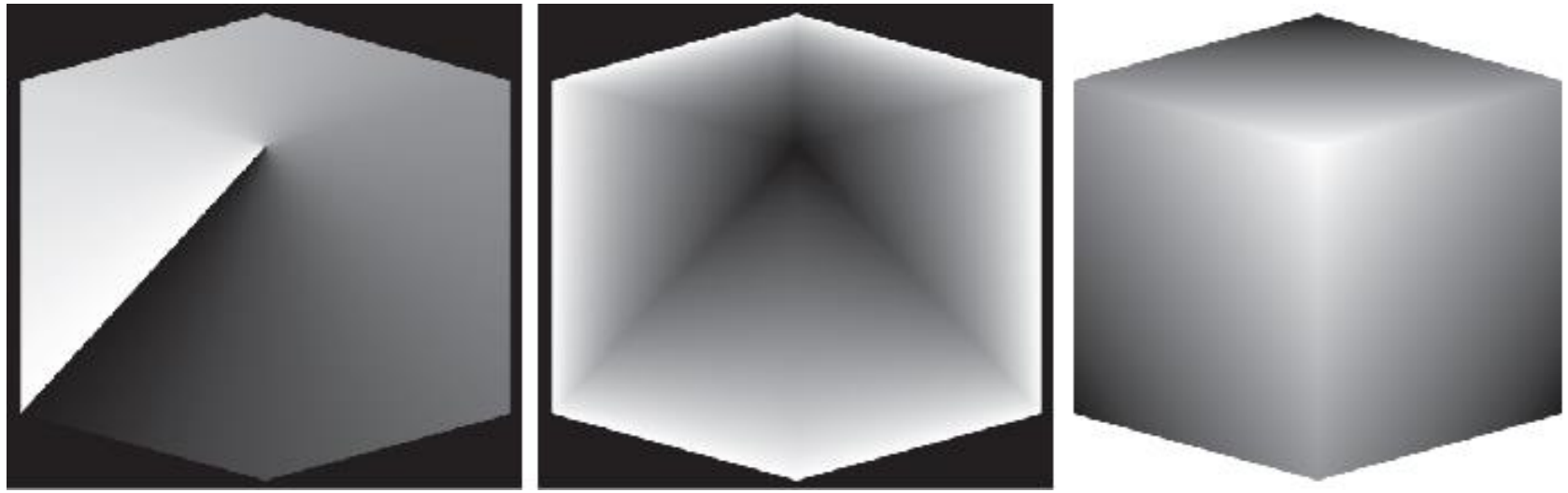



**FIGURE 6.12** The HSI color model based on (a) triangular, and (b) circular color planes. The triangles and circles are perpendicular to the vertical intensity axis.

# HSI Color Model

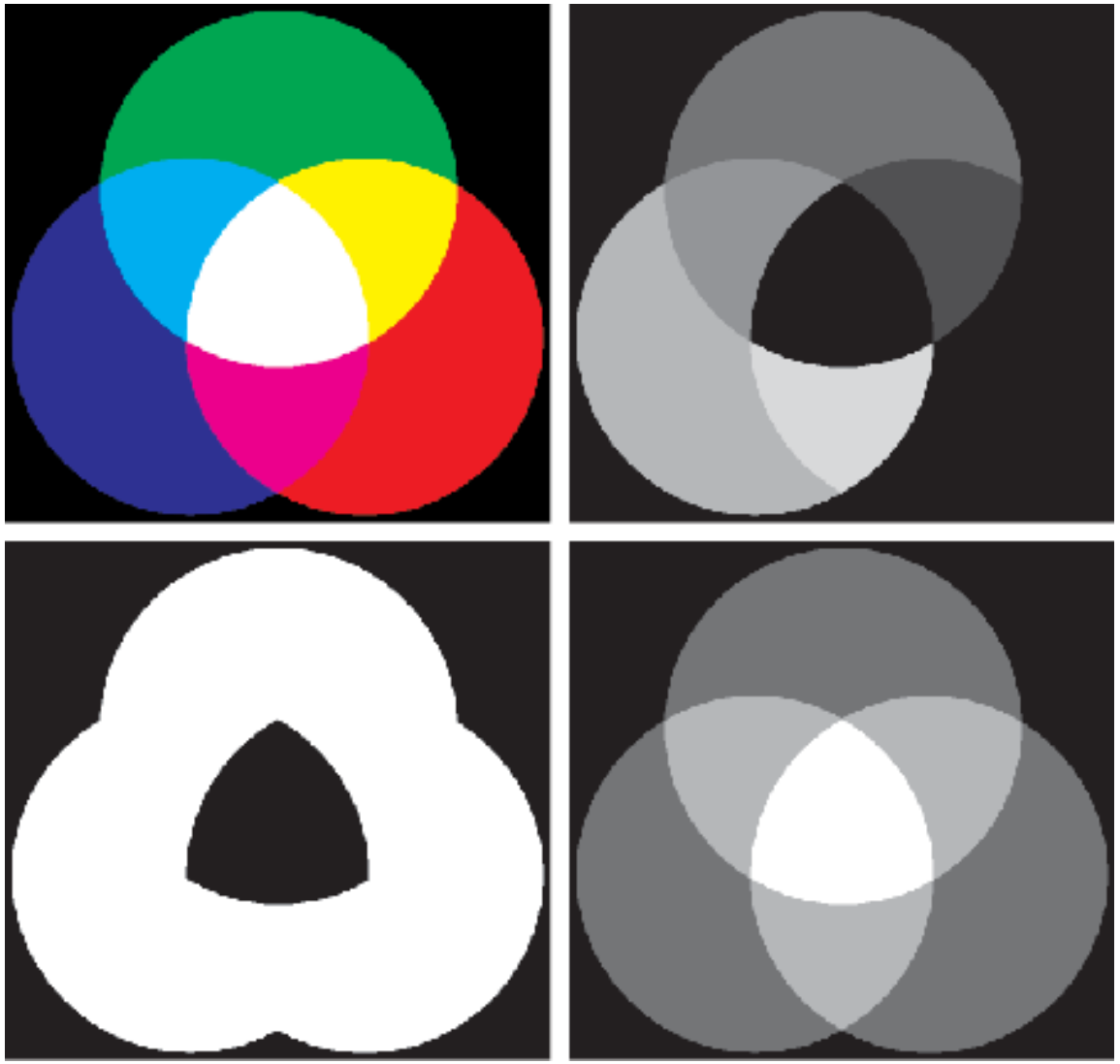




a b c

**FIGURE 6.13** HSI components of the image in Fig. 6.8: (a) hue, (b) saturation, and (c) intensity images.

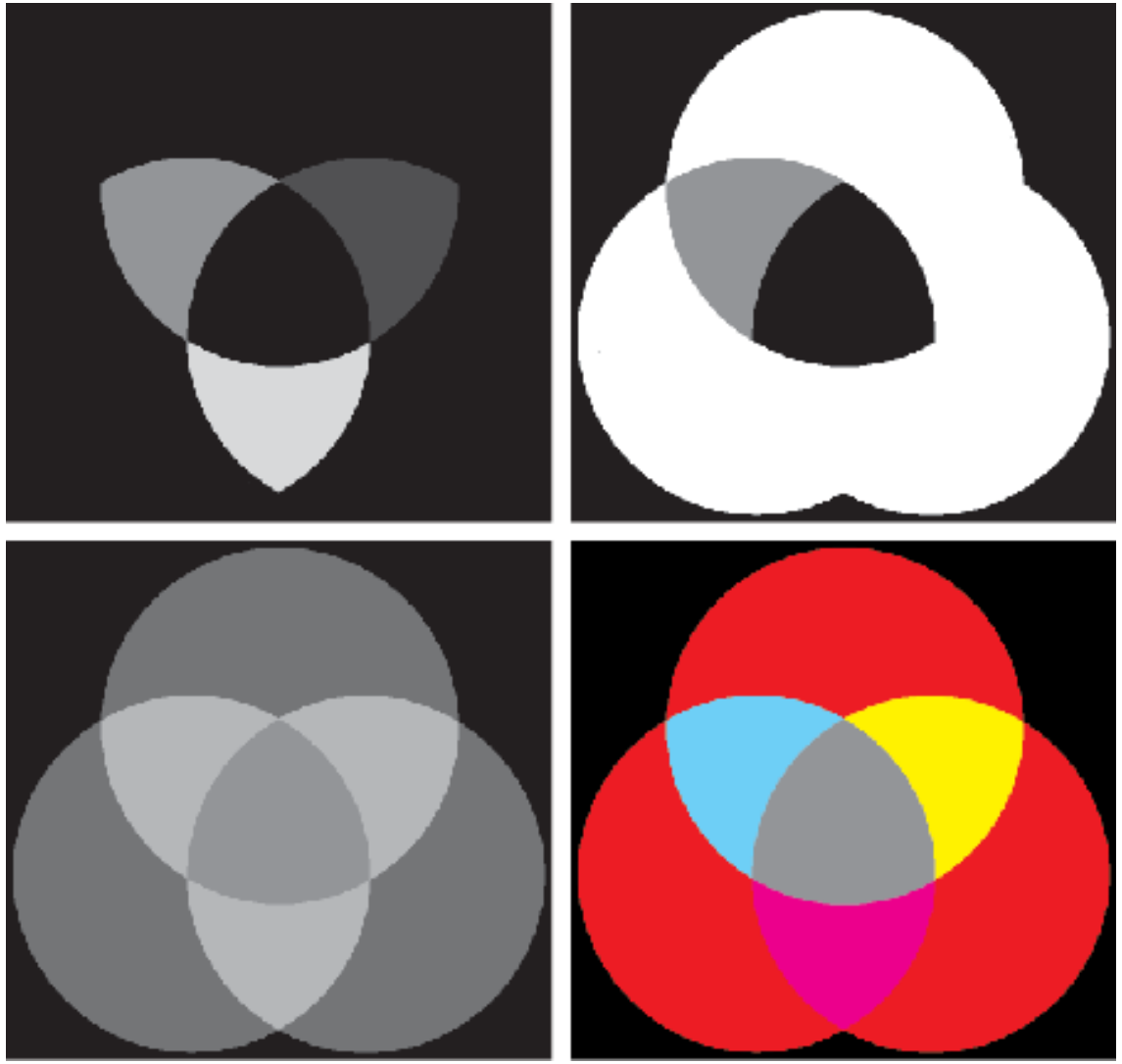
# HSI Color Model



a b
c d

**FIGURE 6.14** (a) RGB image and the components of its corresponding HSI image: (b) hue, (c) saturation, and (d) intensity.

# HSI Color Model



a b
c d

**FIGURE 6.15** (a)-(c) Modified HSI component images. (d) Resulting RGB image. (See Fig. 6.14 for the original HSI images.)

# Pseudo-color Image Processing

- Assign colors to gray values based on a specified criterion

- For human visualization and interpretation of gray-scale events

- Intensity slicing
- Gray level to color transformations

# Intensity Slicing

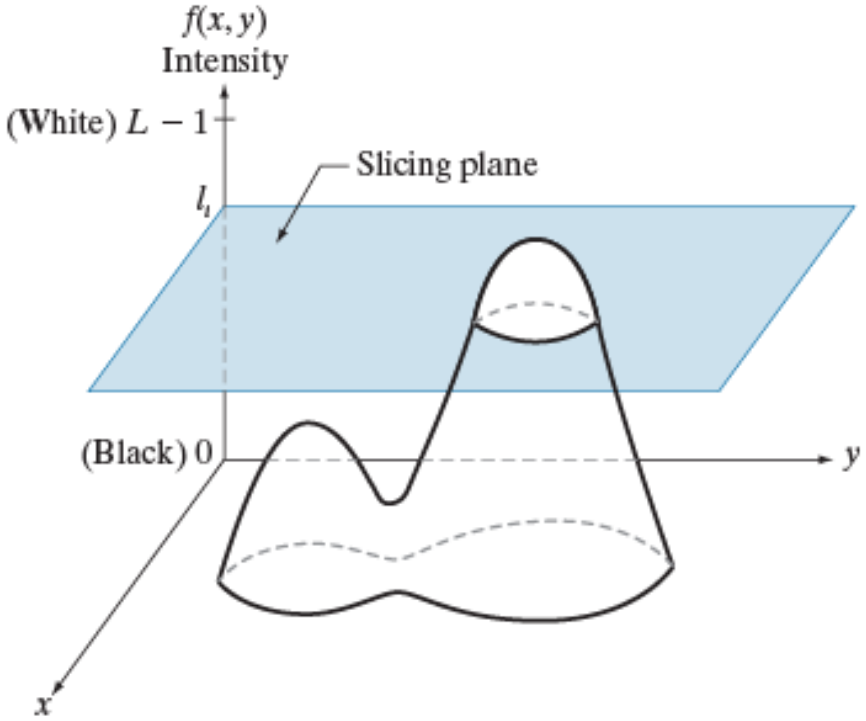



**FIGURE 6.16** Graphical interpretation of the intensity-slicing technique.

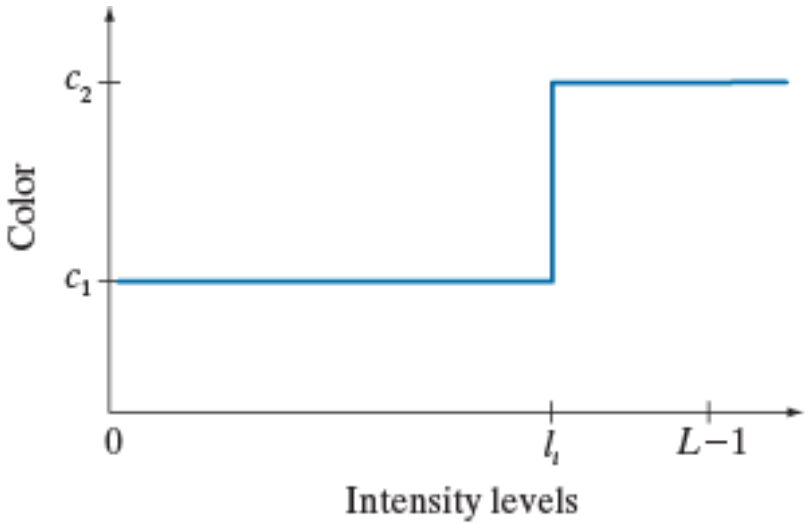
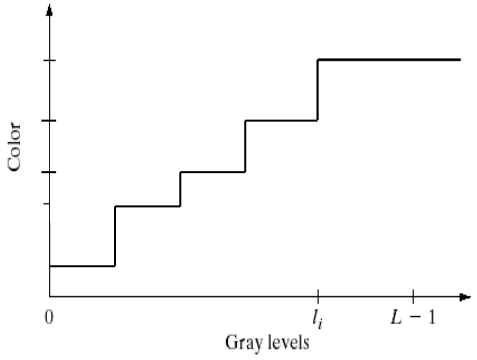
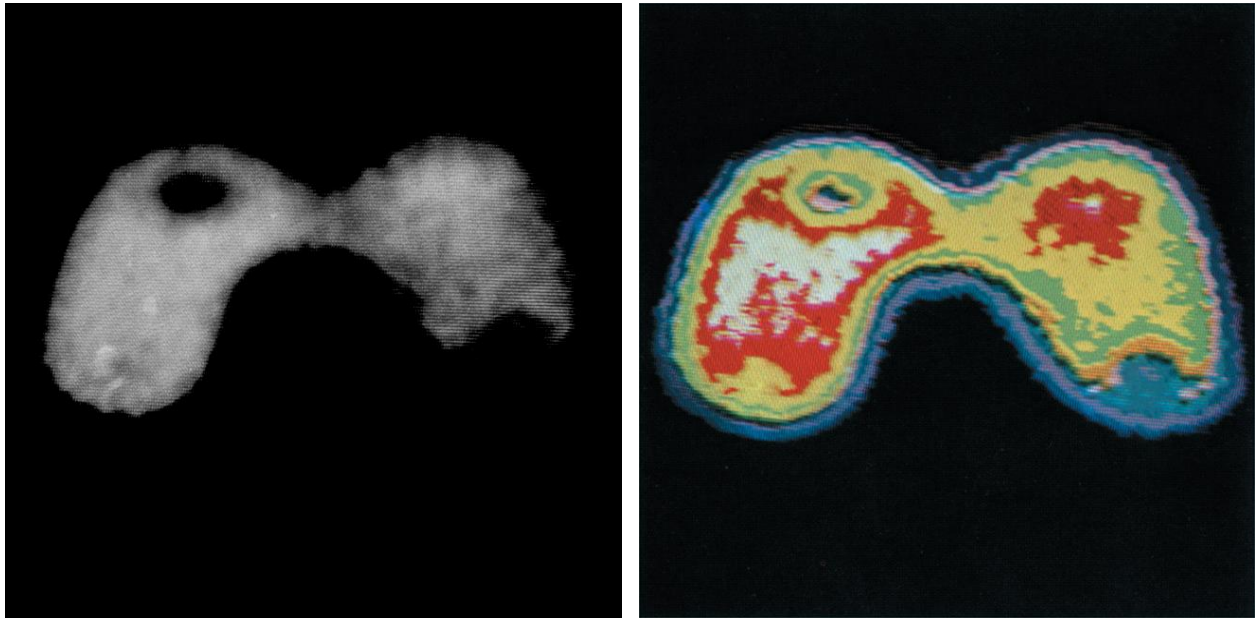



**FIGURE 6.17** An alternative representation of the intensity-slicing technique.

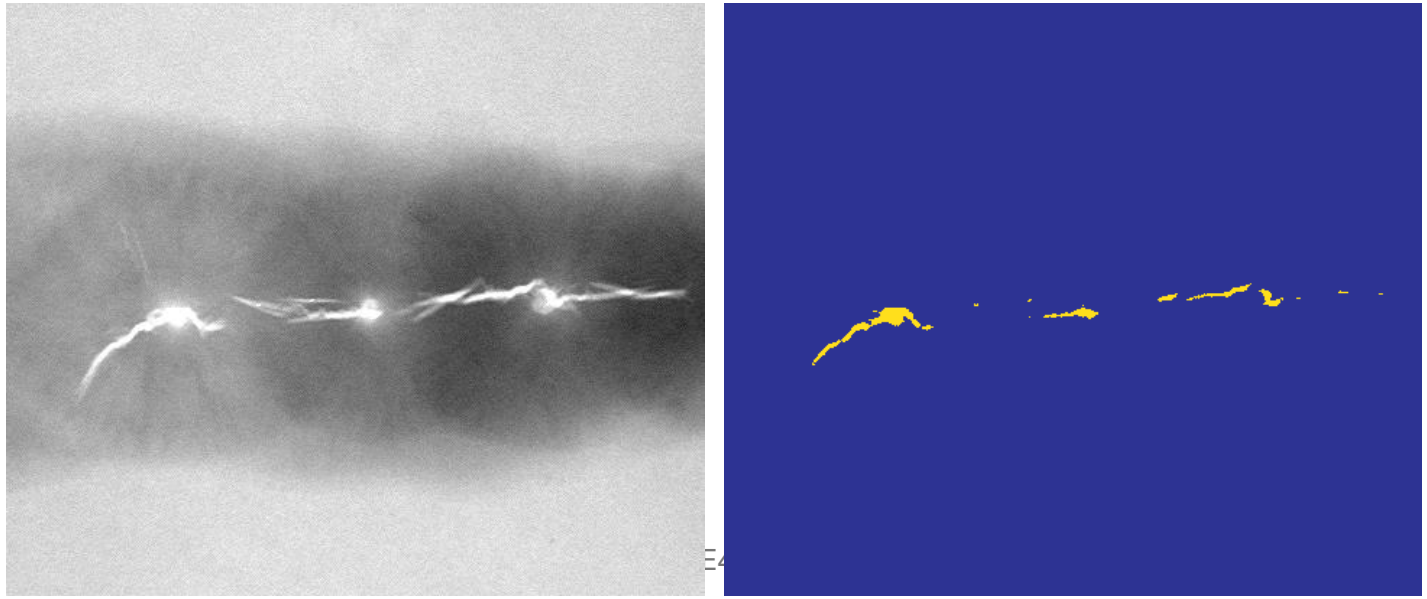
# Intensity Slicing Application



a b

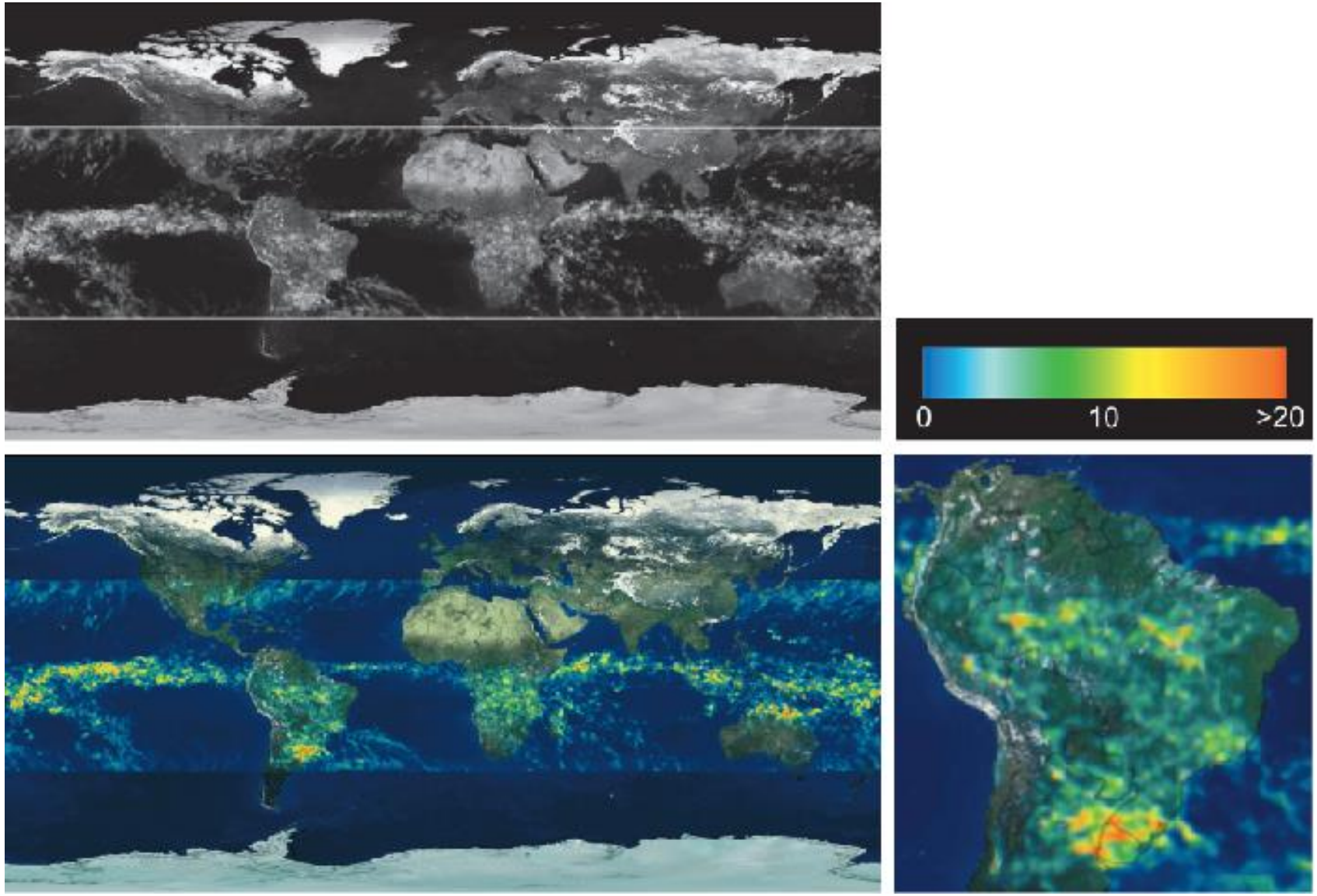
**FIGURE 6.18** (a) Grayscale image of the Picker Thyroid Phantom. (b) Result of intensity slicing using eight colors. (Courtesy of Dr. J. L. Blankenship, Oak Ridge National Laboratory.)



a b

**FIGURE 6.19** (a) X-ray image of a weld. (b) Result of color coding. (Original image courtesy of X-TEK Systems, Ltd.)

# Intensity Slicing Application




a b
c d

**FIGURE 6.20** (a) Grayscale image in which intensity (in the horizontal band shown) corresponds to average monthly rainfall. (b) Colors assigned to intensity values. (c) Color-coded image. (d) Zoom of the South American region. (Courtesy of NASA.)